E0 280: Deep Generative Modeling **Winter 2026**

Lecture 1: Foundations and Hopfield Networks

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1.1 Mc Culloh Pitts Neuron

Simple neuron model can be stated as:

$$n_i(t+1) = \sigma\left(\sum_{j=1}^{n} w_{ji} n_j(t) + b\right) \tag{1.1}$$

and this system continuously evolves.

1.2 Moving to Hopfield

This part is from theory of neural computation book. The problem is store p patterns ξ_i^μ in such a way when presented with new pattern ψ_i, the network outputs the pattern which closely resembles with ψ_i. Patterns are labelled by $\mu = 1, 2, ..., p$ and units in network are labelled by $i = 1, 2, ..., n$. Both $\xi_i^\mu, \psi_i \in \{1, -1\}$.

We want to use neural network to do this. In particular if we start with $S_i = \psi_i$, we want to know if there exists w_{ij} which will make network to go to a state $S_i = \xi_i^{\mu_0}$ where pattern μ_0 has smallest distance with ψ_i.

So dynamics of the network can be written as:

$$S_i = \text{sgn}\left(\sum_{j=1}^{n} w_{ji} S_j - \theta_i\right) \tag{1.2}$$

where $\text{sgn}(x) = 1$ if $x \geq 0$ else -1. For simplicity we drop the bias term and use $S_i = \text{sgn}(\sum_{j=1}^{n} w_{ji} S_j)$. Now we have 2 choices: we can either update all neurons $S(t)$ at a time or we can do one at a time i.e $S_i(t)$. Hence at each time step, we can select a random $S_n(t)$ and it can update itself using above equation.

1.3 Learning One Pattern

Suppose we only have 1 pattern to remember i.e ξ. The condition for this to happen is:

$$S(t+1) = S(t) = \xi \implies S_i(t+1) = S_i(t) = \xi_i \implies \text{sgn}\left(\sum_{j=1}^{n} w_{ji} \xi_j\right) = \xi_i \;\; \forall\, i \tag{1.3}$$

Now we need to choose w_{ij} in a way that this happens. If we choose $w_{ij} \propto \xi_i \xi_j$, this means $w_{ij} = k \xi_i \xi_j$ and the sum becomes $\sum_{j=1}^{n} k \xi_i \xi_j^2$. Since $\xi_i \in \{1, -1\} \implies \xi_j^2 = 1$, hence the sum becomes $\sum_{j=1}^{n} (k \xi_i) = nk\xi_i$.

Therefore, we need to choose $k = \frac{1}{n}$. Now above thing happens if we initialize with $S_i = \xi_i$ and it stops in 1 step.

Now we need to see what happens when we start from arbitrary S_i. Suppose we started with s_i, then:

$$h_i = \sum_{j=1}^{n} w_{ji} s_j = \sum_{j=1}^{n} \frac{1}{n} \xi_i \xi_j s_j = \frac{\xi_i}{n} \sum_{j=1}^{n} \xi_j s_j \tag{1.4}$$

Now s_j can either be ξ_j or $-\xi_j$. Suppose k out of n are wrong, hence $\xi_j s_j = -\xi_j^2 = -1$. For the remaining $n-k$, we will have $\xi_j s_j = \xi_j^2 = 1$. Hence:

$$\sum_{j=1}^{n} \xi_j s_j = 1(n-k) + (-1)k = n - 2k \tag{1.5}$$

Thus, $h_i = \frac{\xi_i}{n}(n-2k) = \xi_i(1 - \frac{2k}{n})$. If $k < \frac{n}{2}$, then h_i will have same sign as ξ_i hence S_i will update itself to ξ_i even if it started with the wrong value. We can also see if $k > \frac{n}{2}$, then h_i will have opposite sign as ξ_i hence S_i will update itself to $-\xi_i$.

1.4 Learning Multiple Patterns

Suppose we now want to store p patterns ξ_i^μ, where $\mu = 1, 2, ..., p$. The extension to the previous case is $w_{ij} = \frac{1}{n} \sum_{\mu=1}^{p} \xi_i^\mu \xi_j^\mu$. If we start with one of the p patterns say ξ_i^n where $n \in \{1, 2, ..., p\}$, the local field h_i is:

$$\begin{aligned} h_i &= \sum_{j=1}^{n} w_{ji} \xi_j^n = \sum_{j=1}^{n} \frac{1}{n} \sum_{\mu=1}^{p} \xi_j^\mu \xi_i^\mu \xi_j^n \\ &= \frac{1}{n} \sum_{\mu=1}^{p} \xi_i^\mu \sum_{j=1}^{n} \xi_j^\mu \xi_j^n \end{aligned}$$

When $\mu = n$, $\sum_{j=1}^{n} \xi_j^\mu \xi_j^n = \sum_{j=1}^{n} (\xi_j^n)^2 = n$, thus:

$$\boxed{h_i = \xi_i^n + \frac{1}{n} \sum_{\mu \neq n}^{p} \xi_i^\mu \sum_{j=1}^{n} \xi_j^\mu \xi_j^n} \tag{1.6}$$

If the second term is small enough, h_i will have the same sign as ξ_i^n. We note that $\mathbb{E}[\xi_j^\mu \xi_j^n] = 0$ and $\mathrm{Var}(\sum_{j=1}^{n} \xi_j^\mu \xi_j^n) = n$. By the Central Limit Theorem, $\sum_{j=1}^{n} \xi_j^\mu \xi_j^n \sim \mathcal{N}(0, n)$. The crosstalk term is $\sim \mathcal{N}(0, p/n)$.

For arbitrary S_i with k errors relative to ξ_i^n:

$$\boxed{h_i = \xi_i^n \left(1 - \frac{2k}{n}\right) + \frac{1}{n} \sum_{\mu \neq n}^{p} \xi_i^\mu \sum_{j=1}^{n} \xi_j^\mu S_j} \tag{1.7}$$

1.5 Proof of Convergence in Serial Mode

A neural network is uniquely defined by $N = (W, T)$. W is symmetric with non-negative diagonal ($W_{ij} = W_{ji}, W_{ii} \geq 0$). Update rule: $s_i(t+1) = \mathrm{sgn}(h_i(t) - T_i)$.

The Energy function is defined as:

$$\boxed{E(t) = s(t)^T W s(t) - 2s(t)^T T = \sum_{i,j} W_{ij} s_i(t) s_j(t) - 2 \sum_i T_i s_i(t)} \tag{1.8}$$

Let $\Delta E = E(t+1) - E(t)$ and $\Delta s_k = s_k(t+1) - s_k(t)$.

$$\begin{aligned}
\Delta E &= (s + \Delta s_k e_k)^T W (s + \Delta s_k e_k) - 2(s + \Delta s_k e_k)^T T - (s^T W s - 2s^T T) \\
&= 2\Delta s_k \left(\sum_{j=1}^{n} W_{kj} s_j(t) - T_k \right) + \Delta s_k^2 W_{kk} \\
&= 2\Delta s_k (h_k(t) - T_k) + \Delta s_k^2 W_{kk} \\
&= 2\Delta s_k H_k(t) + \Delta s_k^2 W_{kk}
\end{aligned}$$

Since Δs_k and $H_k(t)$ always have the same sign (if $\Delta s_k \neq 0$), $2\Delta s_k H_k(t) > 0$. Given $W_{kk} \geq 0$, $\Delta E \geq 0$. Thus, energy increases monotonically (or decreases if defined with a negative sign) and converges as the state space is finite.